

MAGNETIC SIGNALS

You Couldn't Hear—Until Now

What if we could sense the invisible? With next-gen TMR sensors hitting picoTesla sensitivity, healthcare, wearables, and even quantum tech are about to change forever.



KIA NAZARPOUR,
Chief Strategy
Officer at Neuramics,
Professor of
Digital Health at
the University of
Edinburgh, with
experience in
industry R&D, and
co-founder of Rebel
Bionics

Magnetic fields are all around us, yet we rarely notice them. To measure and understand them, we need sensors capable of capturing signals far too weak for the human senses. This is where tunneling magnetoresistance (TMR) sensors step in. By using the quantum tunneling effect between ultra-thin magnetic layers, these sensors translate invisible magnetic fields into measurable electrical signals, giving us a powerful way to detect what was once out of reach.

We have reached a breakthrough with TMR sensors, with a 1000x improvement in sensitivity that allows us to detect magnetic fields at the picoTesla scale—a range once inaccessible to mainstream technology. This advancement means our ability to measure magnetic fields is no longer

confined to traditional uses, unlocking opportunities in healthcare, wearable devices, and quantum computing.

When discussing the achievement of this level of performance, two key design considerations emerge. The first is size. If the sensors are not small enough, they cannot be fitted into all the places we want, whether in consumer gadgets, medical devices, or other portable technologies. The second is power. For wearables and portable devices, every bit of energy matters, unlike in cars, where power is less of an issue. Keeping the sensors both small and energy-efficient is what makes next-generation TMR sensors truly practical.

What can TMR sense next?

Think about gesture recognition. Imagine placing tiny TMR sensors on your skin.

As you move your hand or flex a muscle, those movements generate magnetic signals. The sensors pick them up, and with the help of machine learning, we can map those signals to specific gestures. What is even more exciting is that AI filters out the background noise, leaving us with only the signals that matter. Suddenly, we are talking about powerful new ways to build wearable human-machine interfaces.

In healthcare, TMR sensors have mostly been used as switches, for example, in continuous glucose monitoring. But

